

1-1

```
E94116106@raspberrypi:~ $ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=8.69 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=8.91 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=9.07 ms
^C
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 8.691/8.889/9.066/0.153 ms
E94116106@raspberrypi:~ $
```

1-2

```
E94116106@raspberrypi:~ $ dig google.com

; <<>> DiG 9.16.33-Raspbian <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 25169
;; flags: qr rd ad; QUERY: 1, ANSWER: 9, AUTHORITY: 0, ADDITIONAL: 0
;; WARNING: recursion requested but not available

;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 0       IN      A      142.251.42.238
ns1.google.com.             0       IN      A      216.239.32.10
ns1.google.com.             0       IN      AAAA   2001:4860:4802:32::a
ns2.google.com.             0       IN      A      216.239.34.10
ns2.google.com.             0       IN      AAAA   2001:4860:4802:34::a
ns3.google.com.             0       IN      A      216.239.36.10
ns3.google.com.             0       IN      AAAA   2001:4860:4802:36::a
ns4.google.com.             0       IN      A      216.239.38.10
ns4.google.com.             0       IN      AAAA   2001:4860:4802:38::a

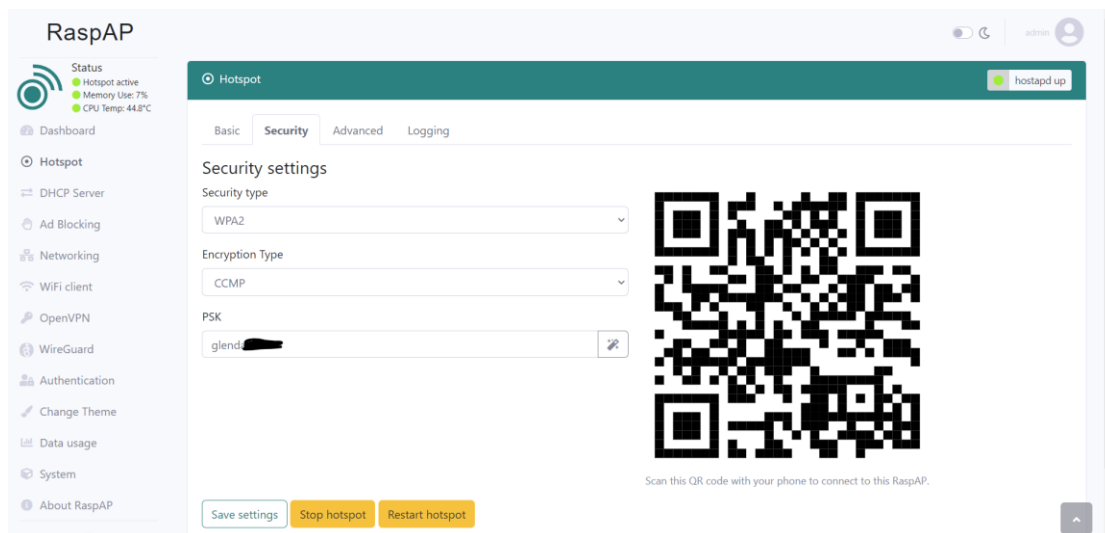
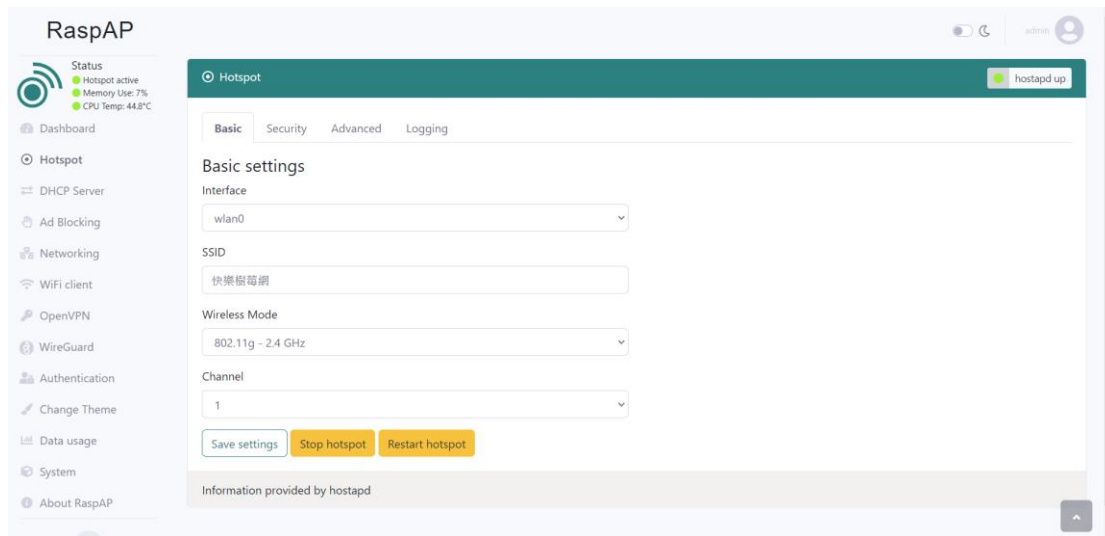
;; Query time: 9 msec
;; SERVER: 192.168.137.1#53(192.168.137.1)
;; WHEN: Tue Nov 29 08:59:49 GMT 2022
;; MSG SIZE rcvd: 286

E94116106@raspberrypi:~ $
```

1-3

```
E94116106@raspberrypi:~ $ traceroute google.com
traceroute to google.com (216.239.32.10), 30 hops max, 60 byte packets
 1 glenn.mshome.net (192.168.137.1) 2.206 ms * *
 2 * * *
 3 140.116.108.253 (140.116.108.253) 2.750 ms 2.664 ms 2.867 ms
 4 140.116.243.33 (140.116.243.33) 2.384 ms 2.683 ms 2.603 ms
 5 140.116.243.193 (140.116.243.193) 2.803 ms 2.527 ms 2.727 ms
 6 140.116.243.177 (140.116.243.177) 3.864 ms 4.739 ms 4.685 ms
 7 192.192.61.146 (192.192.61.146) 4.062 ms 4.258 ms 4.183 ms
 8 192.192.61.24 (192.192.61.24) 7.587 ms 192.192.61.32 (192.192.61.32) 6.797 ms 7.190 ms
 9 192.192.61.184 (192.192.61.184) 8.999 ms 8.845 ms 192.192.61.188 (192.192.61.188) 8.425 ms
10 192.192.61.203 (192.192.61.203) 20.171 ms 20.362 ms 17.510 ms
11 72.14.202.60 (72.14.202.60) 8.600 ms 8.177 ms 8.386 ms
12 * * *
13 108.170.225.176 (108.170.225.176) 8.575 ms 209.85.243.196 (209.85.243.196) 8.791 ms 8.281 ms
14 108.170.244.77 (108.170.244.77) 8.486 ms 8.752 ms 108.170.244.45 (108.170.244.45) 8.633 ms
15 72.14.239.101 (72.14.239.101) 9.170 ms 142.250.224.241 (142.250.224.241) 9.166 ms 72.14.239.101 (72.14.239.101) 9.367 ms
16 142.250.225.150 (142.250.225.150) 12.170 ms 216.239.40.192 (216.239.40.192) 12.397 ms 216.239.41.106 (216.239.41.106) 12.409 ms
17 142.251.66.100 (142.251.66.100) 61.830 ms 142.251.241.118 (142.251.241.118) 57.111 ms 216.239.56.164 (216.239.56.164) 61.596 ms
18 209.85.245.232 (209.85.245.232) 56.602 ms 209.85.242.12 (209.85.242.12) 67.361 ms 142.251.230.148 (142.251.230.148) 58.913 ms
19 108.170.234.59 (108.170.234.59) 60.883 ms 108.170.234.57 (108.170.234.57) 60.994 ms 172.253.68.221 (172.253.68.221) 56.161 ms
20 * * *
21 * * *
22 * * *
23 * * *
24 * * *
25 * * *
26 * * *
27 * * *
28 * * *
29 ns1.google.com (216.239.32.10) 59.607 ms * 56.057 ms
E94116106@raspberrypi:~ $
```

2-1



2-2



2-3

```
E94116106@raspberrypi:~ $ ping 10.3.141.61
PING 10.3.141.61 (10.3.141.61) 56(84) bytes of data.
64 bytes from 10.3.141.61: icmp_seq=1 ttl=64 time=3.48 ms
64 bytes from 10.3.141.61: icmp_seq=2 ttl=64 time=27.2 ms
64 bytes from 10.3.141.61: icmp_seq=3 ttl=64 time=3.39 ms
64 bytes from 10.3.141.61: icmp_seq=4 ttl=64 time=5.40 ms
64 bytes from 10.3.141.61: icmp_seq=5 ttl=64 time=3.01 ms
64 bytes from 10.3.141.61: icmp_seq=6 ttl=64 time=3.43 ms
64 bytes from 10.3.141.61: icmp_seq=7 ttl=64 time=2.89 ms
^C
--- 10.3.141.61 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6009ms
rtt min/avg/max/mdev = 2.889/6.977/27.231/8.304 ms
E94116106@raspberrypi:~ $
```

2-4

DHCP Server

dnsmasq up

Server settings

Advanced

Static Leases

Client list

Logging

Static leases

Clients with a particular hardware MAC address can always be allocated the same IP address. This option adds `dhcp-host` entries to the dnsmasq configuration.

38:d5:7a:66:1a:19

10.3.141.194

glenn

MAC address

IP address

Optional comment

Restrict access

☒ Limit network access to static clients

Enable this option if you want RaspAP to **ignore any clients** which are not specified in the static leases list. This option adds `dhcp-ignore` to the dnsmasq configuration.

Save settings

Stop dnsmasq

Information provided by Dnsmasq

2-5

```
選擇 命令提示字元
C:\Users\glend>tracert google.com
在 上限 30 個躍點上
追蹤 google.com [172.217.160.110] 的路由:
 1  1 ms    1 ms    8 ms    10.3.141.1
 2 143 ms   1 ms    2 ms    140.116.108.253
 3 112 ms   6 ms    2 ms    140.116.243.33
 4 119 ms   3 ms    3 ms    140.116.243.193
 5  98 ms   3 ms    3 ms    140.116.243.177
 6 140 ms   3 ms    3 ms    192.192.61.146
 7 106 ms  12 ms   9 ms    192.192.61.24
 8 130 ms   7 ms   10 ms   192.192.61.184
 9 144 ms   9 ms   7 ms    192.192.61.203
10 141 ms   7 ms   8 ms    72.14.202.60
11 216 ms   9 ms   9 ms    72.14.233.233
12 128 ms  11 ms  227 ms  216.239.48.135
13  80 ms   9 ms   9 ms    tsa03s06-in-f14.1e100.net [172.217.160.110]
追蹤完成。
C:\Users\glend>
```